

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built distributed **search and retrieval** infrastructure on **Elasticsearch** with knowledge-graph enrichment, behind typed REST APIs at **sub-200ms p99**.
- Designed metrics and ran **A/B tests** to validate model changes before full rollout, with a deployment-gated **evaluation framework** tracking quality and regressions.
- Built **LLM retrieval and generation** workflows with **LangGraph** and a **Pinecone** embeddings layer, monitoring **quality, latency, and cost** in production.
- Fine-tuned models with **PyTorch** on multi-GPU clusters and built **vLLM** inference observability with Prometheus/Grafana.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Improved **retrieval relevance**, raising **top-3 recall from 61% to 89%** on a 2K-query eval set with **hybrid dense-sparse retrieval** and **re-ranking**.
- Built **PySpark** jobs for large-scale **feature** and dataset preparation feeding **training, inference, and eval pipelines**.
- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** by moving model inference to an **async, Kubernetes-autoscaled** service.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Integrated **ML models** (scikit-learn, TensorFlow) into production **Java** and **Python** services for **1M+ users** at **99.9% uptime**.
- Improved computer-vision model accuracy **9.6 points** and cut false negatives **31%** through architecture evaluation, feature preprocessing, and threshold tuning.
- Built **SQL** data pipelines processing **50K+ records** with quality control, cutting batch processing from **8 minutes to 90 seconds**.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- **LangGraph** agent with vector retrieval and hybrid re-ranking that decomposes questions into dependency-ordered **SQL** subqueries with cited answers.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- **LLM output evaluation** platform with rubric scoring, **hallucination detection**, and regression alerting on **FastAPI** and **PostgreSQL**.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- **LangGraph** multi-agent system that routes production alerts to specialized sub-agents and synthesizes root-cause reports.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: Python, Java, SQL, TypeScript

ML & Search: PyTorch, TensorFlow, scikit-learn, Ranking, Re-ranking, Elasticsearch, Embeddings, RAG

Experimentation: A/B Testing, Metric Design, Model Evaluation, Feature Engineering, LLM Output Evaluation

Data & Platform: PySpark, Airflow, PostgreSQL, Kubernetes, Docker, AWS, GCP

Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems